

Practical Week -1

Task 1: Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

1. Write a C code to create a process using fork() system call.
2. Read the user input as shell command.
3. Use a system call exec() to execute the shell command.
4. Use a wait() system call for waiting a parent process.
5. Display the PIDs of parent and child processes.

Example Output:

Enter a Linux command: ls

Parent Process

Child Process

Parent PID : 67

Child PID : 68

Child PID : 68

Parent PID : 67

Executing command: ls

Child process completed.

Sol:

```
mohana@MOHANA: ~/command_executor.c

GNU nano 8.7.1      command_executor.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    char command[100];
    pid_t pid;

    printf("Enter a Linux command: ");
    scanf("%s", command);

    pid = fork();

    if (pid < 0)
    {
        printf("Fork failed\n");
        return 1;
    }

    else if (pid == 0)
    {
        // Child process

        printf("Child Process\n");
        printf("Child PID : %d\n", getpid());
        printf("Parent PID : %d\n", getppid());

        printf("Executing command: %s\n", command);

        execlp(command, command, NULL);
    }

    // Parent process
    wait(&pid);
    printf("Parent PID : %d\n", getpid());
    printf("Child PID : %d\n", pid);
    printf("Child process completed.\n");
}
```

```
mohana@MOHANA: ~/command_executor.c

GNU nano 8.7.1      command_executor.c *
}
```

```

else if (pid == 0)
{
    // Child process

    printf("Child Process\n");
    printf("Child PID : %d\n", getpid());
    printf("Parent PID : %d\n", getppid());

    printf("Executing command: %s\n", command);

    execlp(command, command, NULL);

    perror("exec failed");
    exit(1);
}

else
{
    // Parent process

    printf("Parent Process\n");
    printf("Parent PID : %d\n", getpid());

    wait(NULL);

    printf("Child process completed.\n");
}

return 0;
}

```

^G Help

^K Cut

^R Read File

^J Justify

^O Write Out

^T Execute

^N Replace

^F Where Is

^X Exit

^U Paste

mohana@MOHANA: ~

```

mohana@MOHANA:~$ ls
File1.txt      ecec_demo.c      fork_exec.c      task2.c
File2.txt      exec_demo        namedirectory    task3
OSSP           exec_demo.c      skill4           task3.c
OS_LAB         filename         task1
command_executor.c.save filename.c        task1.c
ecec_demo      fork_exec        task2

mohana@MOHANA:~$ nano command_executor.c
mohana@MOHANA:~$ gcc command_executor.c -o command_executor
mohana@MOHANA:~$ ./command_executor
Enter a Linux command: ls
Parent Process
Parent PID : 550
Child Process
Child PID : 551
Parent PID : 550
Executing command: ls
File1.txt      ecec_demo.c      skill4
File2.txt      exec_demo        task1
OSSP           exec_demo.c      task1.c
OS_LAB         filename         task2
command_executor filename.c        task2.c
command_executor.c fork_exec        task3
command_executor.c.save fork_exec.c      task3.c
ecec_demo      namedirectory
Child process completed.
mohana@MOHANA:~$ ./command_executor
Enter a Linux command: pwd
Parent Process
Parent PID : 617
Child Process
Child PID : 620
Parent PID : 617
Executing command: pwd
/home/mohana
Child process completed.
mohana@MOHANA:~$

```

Task-2: Using Linux terminal commands (uname, lscpu, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

Sol:

Command	What it shows

uname	Linux/system information
uname -a	Detailed kernel/system information
lscpu	CPU architecture and processor information
ps	Current processes
ps aux	Detailed process information
top	Real-time process and resource usage

mohana@MOHANA: ~

```
mohana@MOHANA:~$ uname
Linux
mohana@MOHANA:~$ uname -a
Linux MOHANA 6.18.33.2-microsoft-standard-WSL2 #1 SMP PRE
EMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Li
mohana@MOHANA:~$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          39 bits physical, 48 bits vi
Byte Order:             Little Endian
CPU(s):                 12
On-line CPU(s) list:    0-11
Vendor ID:              GenuineIntel
Model name:             13th Gen Intel(R) Core(TM) i
CPU family:             6
Model:                  186
Thread(s) per core:     2
Core(s) per socket:     6
Socket(s):              1
Stepping:               3
BogoMIPS:               4991.99
Flags:                  fpu vme de pse tsc msr pae m
                        apic sep mtrr pge mca cmov p
                        6 clflush mmx fxsr sse sse2
                        yscall nx pdpe1gb rdtscp lm
                        t_tsc rep_good nopl xtopolog
                        eliable nonstop_tsc cpuid ts
                        _freq pni pclmulqdq vmx ssse
                        x16 pcid sse4_1 sse4_2 x2api
                        popcnt tsc_deadline_timer a
                        e avx f16c rdrand hypervisor
                        m abm 3dnowprefetch ssbd ibr
                        stibp ibrs_enhanced tpr_shad
                        vpid ept_ad fsgsbase tsc_adj
                        1 avx2 smep bmi2 erms invpci
                        d adx smap clflushopt clwb s
                        saveopt xsavec xgetbv1 xsave
```

mohana@MOHANA: ~

```
mohana@MOHANA:~$ ps
  PID TTY          TIME CMD
   319 pts/0        00:00:00 bash
   733 pts/0        00:00:00 ps
mohana@MOHANA:~$ ps aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TI
ME COMMAND
mohana      1  0.3  0.3  24328 15372 ?        Ss   05:20   0:
03 /sbin/init
mohana      2  0.0  0.0    3180  2208 hvc0     Sl+  05:20   0:
00 /init
mohana      7  0.0  0.0    3180  2048 hvc0     Sl+  05:20   0:
00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-f
mohana     53  0.0  0.4  50396 16804 ?        S<s  05:20   0:
00 /usr/lib/systemd/systemd-journald
systemd+   70  0.0  0.3  22416 14468 ?        Ss   05:20   0:
00 /usr/lib/systemd/systemd-resolved
mohana     92  0.1  0.3  35140 12256 ?        Ss   05:20   0:
01 /usr/lib/systemd/systemd-udevd
mohana    153  0.0  0.0    2888  1948 ?        Ss   05:20   0:
00 /bin/sh /usr/lib/systemd/scripts/chronyd-starter.sh -n -F 1
mohana    154  0.0  0.0    4472  3100 ?        Ss   05:20   0:
00 /usr/sbin/cron -f -P
message+   155  0.0  0.1    8820  5316 ?        Ss   05:20   0:
00 @dbus-daemon --system --address=systemd: --nofork --nopidfile
mohana    159  0.0  0.7  44096 29712 ?        Ss   05:20   0:
00 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-t
mohana    168  0.0  0.2  18436  9388 ?        Ss   05:20   0:
00 /usr/lib/systemd/systemd-logind
syslog    175  0.0  0.1 220548  5568 ?        Ssl  05:20   0:
00 /usr/sbin/rsyslogd -n -iNONE
```

```
mohana      216  0.0  0.0   5492  2660 tty1      Ss+  05:20  0:
00 /usr/sbin/agetty --noret --noclear --issue-file=/etc/issue:
mohana      225  0.0  0.8 123460 32772 ?        Ssl  05:20  0:
00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-up
_chrony      226  0.0  0.2  23588 11084 ?        S    05:20  0:
00 /usr/sbin/chronyd -n -F 1 -x
```

mohana@MOHANA: ~

mohana@MOHANA:~\$ top

```
top - 05:43:42 up 2:51, 1 user, load average: 0.09, 0.04, 0.
Tasks: 24 total, 1 running, 23 sleeping, 0 stopped, 0 z
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi,
MiB Mem : 3758.6 total, 3048.4 free, 473.1 used, 312.
MiB Swap: 1024.0 total, 1024.0 free, 0.0 used. 3285.
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM
1	mohana	20	0	24328	15372	11568	S	0.0	0.4
2	mohana	20	0	3180	2208	2072	S	0.0	0.1
7	mohana	20	0	3180	2048	1940	S	0.0	0.1
53	mohana	19	-1	50396	16804	15500	S	0.0	0.4
70	systemd+	20	0	22416	14468	11972	S	0.0	0.4
92	mohana	20	0	35140	12256	9184	S	0.0	0.3
153	mohana	20	0	2888	1948	1820	S	0.0	0.1
154	mohana	20	0	4472	3100	2852	S	0.0	0.1
155	message+	20	0	8820	5316	4636	S	0.0	0.1
159	mohana	20	0	44096	29712	14576	S	0.0	0.8
168	mohana	20	0	18436	9388	8144	S	0.0	0.2
175	syslog	20	0	220548	5568	4648	S	0.0	0.1
216	mohana	20	0	5492	2660	2476	S	0.0	0.1
225	mohana	20	0	123460	32772	18164	S	0.0	0.9
226	_chrony	20	0	23588	11084	7128	S	0.0	0.3
237	_chrony	20	0	12096	2452	1808	S	0.0	0.1
316	mohana	20	0	3184	1112	980	S	0.0	0.0
317	mohana	20	0	3200	1248	1108	S	0.0	0.0
319	mohana	20	0	6268	5532	3864	S	0.0	0.1
322	mohana	20	0	8648	5544	4700	S	0.0	0.1
385	mohana	20	0	22340	13604	10976	S	0.0	0.4
387	mohana	20	0	22920	4092	2088	S	0.0	0.1
412	mohana	20	0	6136	5424	3864	S	0.0	0.1
755	mohana	20	0	8164	6064	3948	R	0.0	0.2